

Design Tables - 175mm² ACSR - (LYNX) Normal Altitude

Basic Design Span	90
Max Clashing Span ***	108 (135)
Occasional Long Span (2.0m Phase to Phase)	160
FCD Required	No
Clashing Weather Zone	2C
Windspan factor of safety applied to Wood Poles	2.5
Strut load factor of safety applied to Wood Poles	2.5

In Line Structures

Drawing No	Support Type	Support Class	Pin Type	Windspan Limit
434004	Intermediate	Stout	10kN	114m
434004	Intermediate	E/Stout	10kN	153m
434007/2 *	Intermediate	Stout 'H'	-	156+m
434005	Section	Stout	-	114m
434005	Section	E/Stout	-	153m

Angle Structures

Drawing No.	Support Type	Support Class	Pin Type	Line Deviation	Stays
434004	Pin Angle	Stout	10kN	9°	1 @ 45°
434004	Pin Angle	Stout <=13m	10kN	9°	1 @ 30°
434005	Sect Angle	Stout	-	20°	2 @ 45°
434005	Sect Angle	E/Stout	-	44°	2 @ 45°
434005	Sect Angle	E/Stout <=13m	-	29°	2 @ 30°
434006	H Sect Angle	Stout H <=13m	-	60°	4 @ 45°

Terminal Structures

Drawing No.	Support Type	Support Class	Stays
434011	Terminal	Stout H	4 @ 45°
434011	Terminal	Stout H <=13m	4 @ 40°

- * Used for occasional long spans.
- *** (value in brackets = actual clashing limit for given basic span)
- See requirements on achieving 90° Dev
- Stay requirements are based on the total number of 7/4.00mm Grade 1150 stays per structure.

Notes

This table is for use with lines located in Normal environments ie lines located in altitudes above below 300m sea level.

Due to the higher tension in this design, tension sets shall utilise the 125Kn insulators and fittings as detailed on http://mwapps2/dbd_eeuk/docs/NSP004127.pdf page 24 & 25.

Design Tables - 175mm² ACSR - (Lynx) High Altitude

Basic Design Span	90
Max Clashing Span	108 (125)
Occasional Long Span (2.0m Phase to Phase)	160
FCD Required	No
Weather	3C
Windspan factor of safety applied to Wood Poles	2.0
Strut load factor of safety applied to Wood Poles	2.5

Comment [gmh1]: Only completed this work for normal environment, need to check severe or high up environment.

In Line Structures

Drawing No	Support Type	Support Class	Pin Type	Windspan Limit
434004	Intermediate	Stout	10kN	62m
434004	Intermediate	E/Stout	10kN	85m
434004	Intermediate	12m E/Stout	10kN	94m **
434007/2 *	Intermediate	Stout H	10kN	160m
434005	Section	E/Stout	-	85m
434005	Section	12m E/Stout	-	94m **

Angle Structures

Drawing No.	Support Type	Support Class	Pin Type	Line Deviation	Stays
434004	Pin Angle	Stout	10kN	8°	1 @ 45°
434004	Pin Angle	Stout <=13m	10kN	8°	1 @ 30°
434005	Sect Angle	E/Stout <=14m	-	41°	2 @ 45°
434005	Sect Angle	E/Stout <=13m	-	36°	2 @ 40°
434005	Sect Angle	E/Stout <=13m	-	31°	2 @ 35°
434005	Sect Angle	E/Stout <=13m	-	26°	2 @ 30°
434006	H Sect Angle	Stout H	-	60°	4 @ 45°
434006	H Sect Angle	Stout H <=13m	-	60°	4 @ 40°

Terminal Structures

Drawing No.	Support Type	Support Class	Stays
434011	Terminal	Stout H	4 @ 45°
434011	Terminal	Stout H <=13m	4 @ 40°

- * Used for occasional long spans.
- Stay requirements are based on the total number of 7/4.00mm Grade 1150 stays per structure.
- Angle & terminal structures based on higher basic span

Notes

This table is for use with lines located in severe environments ie lines located in altitudes above 300m above sea level.

Due to the higher tension in this design, tension sets shall utilise the 125Kn insulators and fittings as detailed on http://mwapps2/dbd_eeuk/docs/NSP004127.pdf page 24 & 25.

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	LYNX
Greased Conductor Weight (kg/m)	0.8626
Cross Sectional Area of Conductor (mm²)	226.2
Conductor Overall Diameter (mm)	19.5
Coefficient of Linear Expansion (/Degree C)	1.8E-05
Modulus of Elasticity (kg/mm²)	8157.73
Rated Breaking Strength of Conductor (kgf)	8137.3
Basic / Recommended Span (m)	76.2
Wind Pressure on Conductor (N/m²)	390
Radial Ice Thickness (mm)	9.5
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	2325.0
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	1627.5
Temperature at EDT Limit (Degrees C)	5
Maximum Conductor Tension (MCT) (kgf) at	-5.6°C 2284.5
Maximum Conductor Weight (MCW) (kg/m)	1.653
Maximum Conductor Pressure (MCP) (kg/m)	1.531
Freezing Point Tension (FPT) (kgf) at	0°C 1772.1

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	LYNX
Basic / Recommended Span (m)	76.2
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	8.0 at 15°C
Required Percentage Increase in Tension (%)	8 at 15°C

Temp. (Deg. C)	Tension (kgf)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	2062.5	0.19	0.26	0.33	0.42	0.52	0.63	0.75	0.88	1.69
0	1893.2	0.21	0.28	0.36	0.46	0.57	0.69	0.82	0.96	1.85
5	1745.3	0.22	0.30	0.40	0.50	0.62	0.75	0.89	1.04	2.00
10	1601.5	0.24	0.33	0.43	0.55	0.67	0.81	0.97	1.14	2.18
15	1463.0	0.27	0.36	0.47	0.60	0.74	0.89	1.06	1.25	2.39
20	1331.1	0.29	0.40	0.52	0.66	0.81	0.98	1.17	1.37	2.62
25	1207.2	0.32	0.44	0.57	0.72	0.89	1.08	1.29	1.51	2.89
30	1093.1	0.36	0.48	0.63	0.80	0.99	1.19	1.42	1.67	3.20
35	989.9	0.39	0.53	0.70	0.88	1.09	1.32	1.57	1.84	3.53
40	898.3	0.43	0.59	0.77	0.97	1.20	1.45	1.73	2.03	3.89
45	818.4	0.47	0.65	0.84	1.07	1.32	1.59	1.90	2.23	4.27
50	749.5	0.52	0.70	0.92	1.17	1.44	1.74	2.07	2.43	4.66
55	690.6	0.56	0.77	1.00	1.26	1.56	1.89	2.25	2.64	5.06
60	640.2	0.61	0.83	1.08	1.36	1.68	2.04	2.43	2.85	5.46
65	597.1	0.65	0.88	1.16	1.46	1.81	2.18	2.60	3.05	5.85
70	560.1	0.69	0.94	1.23	1.56	1.93	2.33	2.77	3.25	6.24
75	528.1	0.74	1.00	1.31	1.65	2.04	2.47	2.94	3.45	6.62
80	500.2	0.78	1.06	1.38	1.75	2.16	2.61	3.10	3.64	6.98

This sag chart has been produced to try and mimic the original design sag charts as closely as possible thus minimising out of balance situations were transitions occur with lines built to the original specification.